Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

ClusterMind

Team name: ClusterMind

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Problem: Kubernetes dependencies are difficult to trace during incidents. Engineers need a clearer view of resource health, relationships, and possible impact.

Theme alignment: Future of Work & Enterprise Productivity

Brief about the idea

ClusterMind: an AI powered Kubernetes infrastructure digital twin

ClusterMind reads resource metadata from a reachable Kubernetes API and builds a confidence rated dependency graph. Gemini 3.8 Flash is configured to explain possible impact and suggest what to investigate next.

DISCOVER

Collect resource status, configuration metadata, and events from a reachable Kubernetes API.

CONNECT

Infer relationships and visualize topology with confidence ratings.

INVESTIGATE

Trace potential impact, summarize severity, and suggest evidence based next checks.

For SREs, platform engineers, DevOps teams, developers, and teams onboarding to an unfamiliar cluster.

Opportunities

How is it different? | How does it help? | What is the unique value?

During an incident, teams often need to inspect many Kubernetes resources to understand dependencies. ClusterMind turns that scattered metadata into a map engineers can inspect and reason about.

HOW IT DIFFERS

Interactive topology and deterministic graph impact analysis, with Gemini 3.8 Flash configured for AI explanations.

HOW IT HELPS

Trace possible failure paths, inspect resource health, and prioritize what to check next.

CORE USP

Ask “What could this failure affect?” and see a visual impact map with a human readable explanation.

Potential value: speed up first pass triage, support safer changes, and help teams learn unfamiliar systems. The MVP does not automatically remediate incidents.

Features implemented in the submitted code

- Read access to Namespaces, Deployments, StatefulSets, DaemonSets, Pods, Services, Ingresses, ConfigMaps, HPAs, NetworkPolicies, and Events.
- Namespace resource counts and health rollups. Resource details show status, readiness, metadata, and relationships.
- Dependency edges inferred from Service selectors, Ingress backends, HPA targets, environment references, and name prefixes, with confidence scores.
- Interactive React Flow graph with pan, zoom, minimap, node selection, dependency inspection, and impact highlighting.
- Breadth first impact tracing capped at five hops, with direct and indirect impact paths.
- Gemini 3.8 Flash is configured for severity and impact explanations and structured investigation reports. Evidence references are checked against the supplied timeline.
- CLI, REST API, and server sent events poll every 30 seconds. Kubernetes Secret objects are excluded. ClusterMind does not write to cluster resources.

MVP: map dependencies, assess likely impact, and guide the next check. No automated remediation.

How ClusterMind works

1. CONNECT
Use kubeconfig or access from inside the cluster. Limit collection by namespace.

2. COLLECT
Read permitted resource metadata and events. Kubernetes Secret objects are not collected.

3. BUILD GRAPH
Infer relationships and filter low confidence edges.

4. TRACE IMPACT
BFS traces impact across up to five hops.

5. EXPLAIN
Gemini 3.8 Flash explains results. Evidence is validated.

USE CASE A · UNDERSTAND A SERVICE
Select a node to inspect health, metadata, and upstream or downstream dependencies.

USE CASE B · INVESTIGATE A FAILURE
Choose a target, review its potential impact area, and use suggested checks to continue triage.

Human stays in control: ClusterMind explains and highlights. Engineers verify evidence and choose the response.

Conceptual investigation workflow

ClusterMind | Cluster: demo | Namespace: payments | SAMPLE DATA

TOPOLOGY + HEALTH

Ingress, Service, and Deployment relationships. Inspect resource health, metadata, and dependencies.

SELECT A TARGET

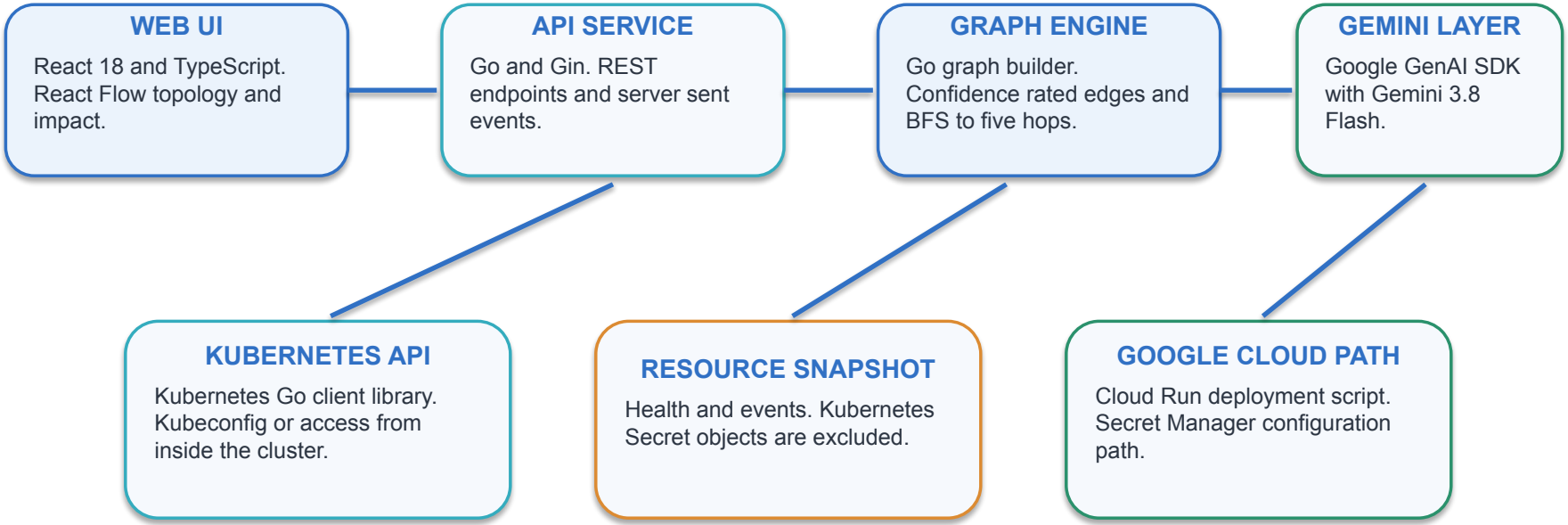
Graph traversal identifies possible impact. Gemini 3.8 Flash explains the result.

IMPACT + NEXT CHECKS

View direct and indirect impact, an AI summary, evidence references, and suggested debugging checks.

Conceptual example with sample names. Live discovery needs Kubernetes API access and RBAC permissions.

Architecture in the submitted code



Cloud Run uses Gemini 3.8 Flash. Live Kubernetes discovery requires network access to the Kubernetes API and appropriate RBAC permissions.

Technology stack and why it is used

Backend	Go + Gin REST server; Kubernetes Go client library
Kubernetes data	Read access to workload, networking, configuration metadata, and Events
Graph & analysis	Confidence rated dependency inference; breadth first impact traversal to depth 5
AI integration	Gemini 3.8 Flash explains severity, possible causes, and next checks. Graph traversal stays deterministic.
Frontend	React 18, TypeScript, React Flow v12, React Query, Zustand, and Tailwind CSS
Delivery	Dockerfile and Cloud Run deployment script; Cloud Build and Artifact Registry path
API KEY STORAGE	GEMINI_API_KEY is configured through environment settings or Secret Manager, not hardcoded in source

Cost and deployment considerations

ClusterMind's operating costs are driven by three components: local development, Gemini API usage, and Google Cloud infrastructure.

LOCAL DEMO

Use Kind or a local Kubernetes cluster for development and testing without managed cluster costs.

GEMINI API

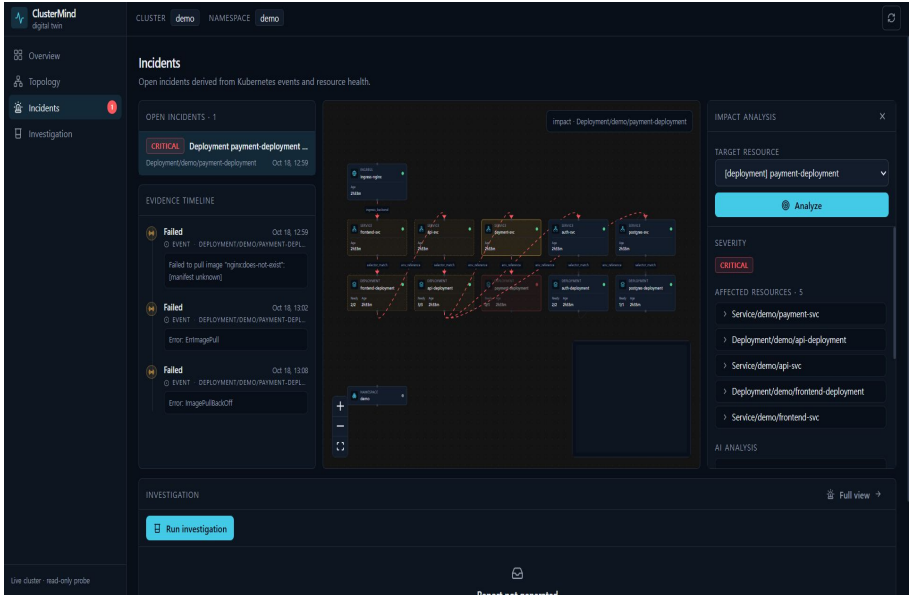
Paid tier through December 31, 2026: $0.75 per million input tokens and $3.75 per million output tokens. Costs scale with token usage.

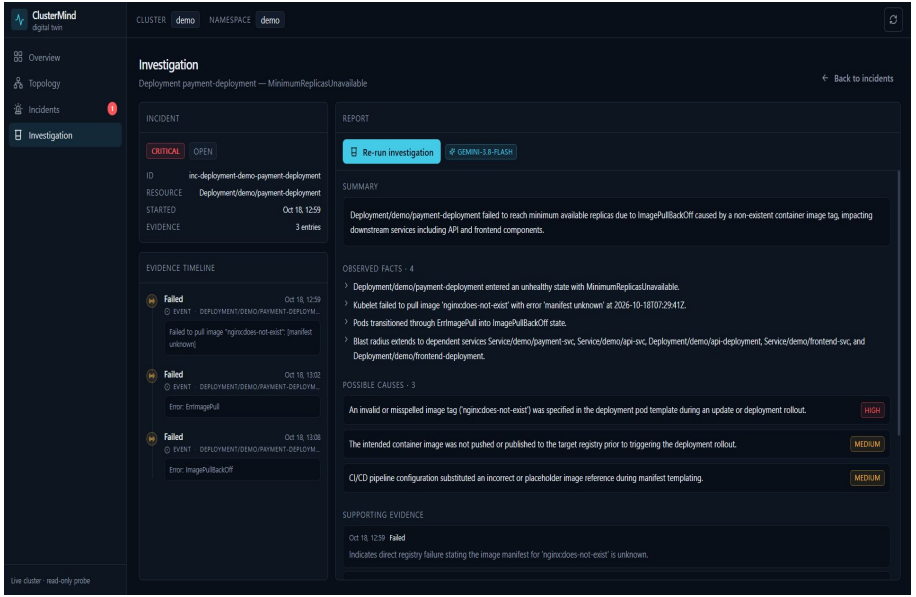
CLOUD RUN + CLUSTER

Cloud Run includes a monthly free tier. Additional compute usage, image storage, logging, networking, and GKE resources may incur charges.

Live Kubernetes discovery requires authenticated network access to the Kubernetes API and appropriate RBAC permissions.

Product screenshots





Blast Radius Analysis
Visualizes affected services and dependency paths when a Kubernetes resource fails.

AI Powered Investigation
Gemini 3.8 Flash summarizes observed evidence, possible causes, and supporting findings.

Validation and performance status

GRAPH RULE 1

Service label selectors map to matching workload labels.

GRAPH RULE 2

Ingress backend references map to Services.

GRAPH RULE 3

Service to Deployment relationships are inferred.

GRAPH RULE 4

Resource node IDs remain stable.

TEST COVERAGE IN THE REPOSITORY

Go tests cover graph rules, impact traversal, investigation handling, and API behavior.

IMPACT TRAVERSAL BENCHMARK

100 nodes · 300 edges	**0.44 ms**
500 nodes · 1,500 edges	**1.39 ms**
1,000 nodes · 3,000 edges	**2.20 ms**

Median of five runs on synthetic graphs. Results exclude Kubernetes API calls and Gemini latency.

Next steps and future development

AI reliability and validation

Automate checks for Gemini response quality, evidence grounding, and fallback behavior.

Expand performance benchmarks and validate impact accuracy against known test topologies.

Deeper cluster intelligence

Replace fixed interval polling with Kubernetes watches or informers.

Correlate Prometheus metrics and logs with Kubernetes Events already collected.

Improve graph layout and expand relationship inference for Network Policies.

Production readiness

Add historical snapshots, topology comparisons, and deployment risk previews.

Strengthen authentication, least privilege access, and credential rotation.

Product principle: explain first, show evidence, and keep operators in control. Automated remediation is outside this MVP.

Submission links

Explore the source code, watch the product demonstration, and open the deployed application.

PUBLIC GITHUB REPOSITORY

Open the public repository

3 MINUTE DEMO VIDEO

Watch the ClusterMind demo on YouTube

FINAL PRODUCT URL

Open the deployed ClusterMind prototype

ClusterMind helps engineers trace potential failure impact, investigate incidents with Gemini, and make informed decisions using evidence from their Kubernetes environment.

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Thank you

